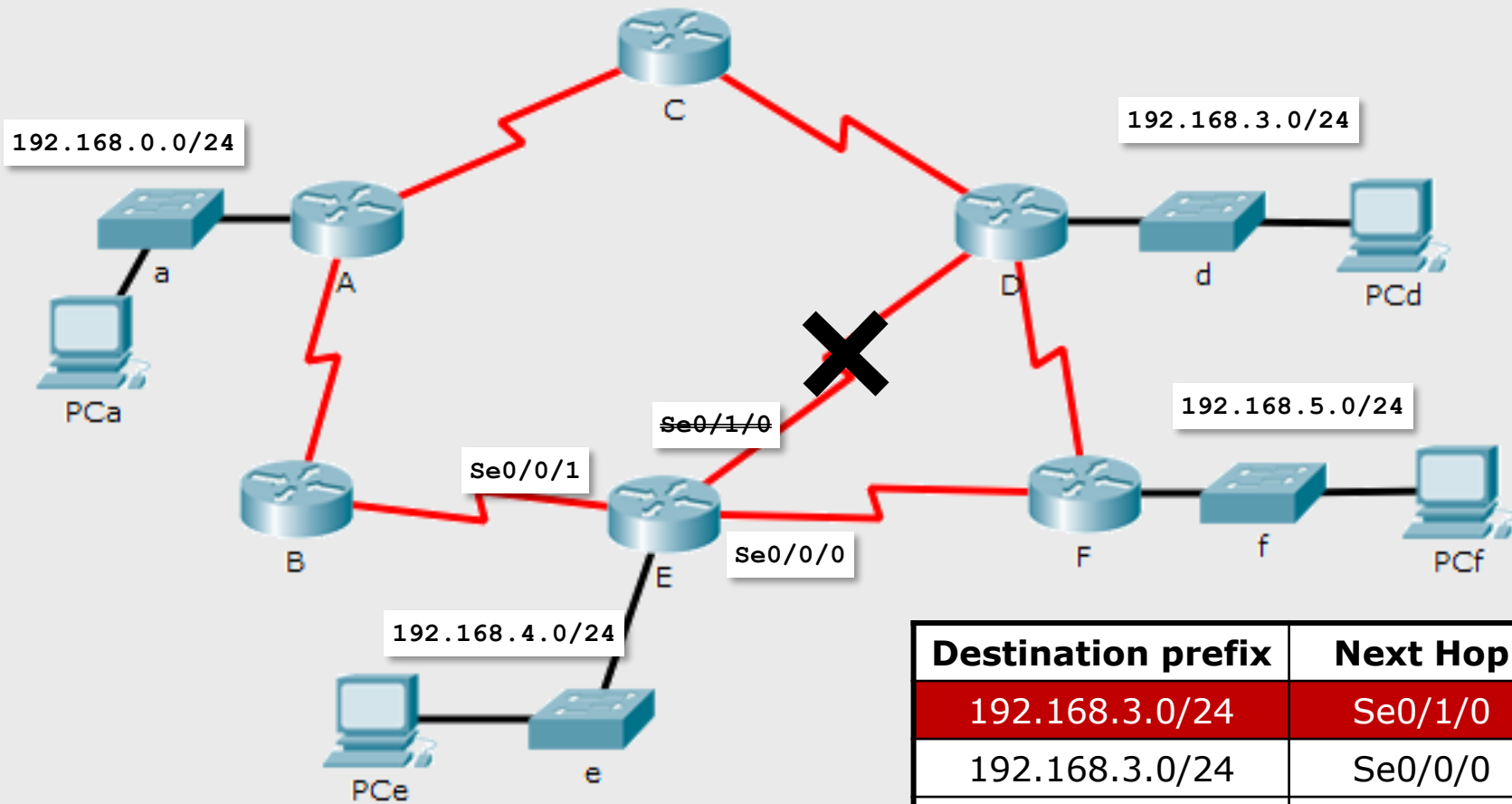


Lab 6

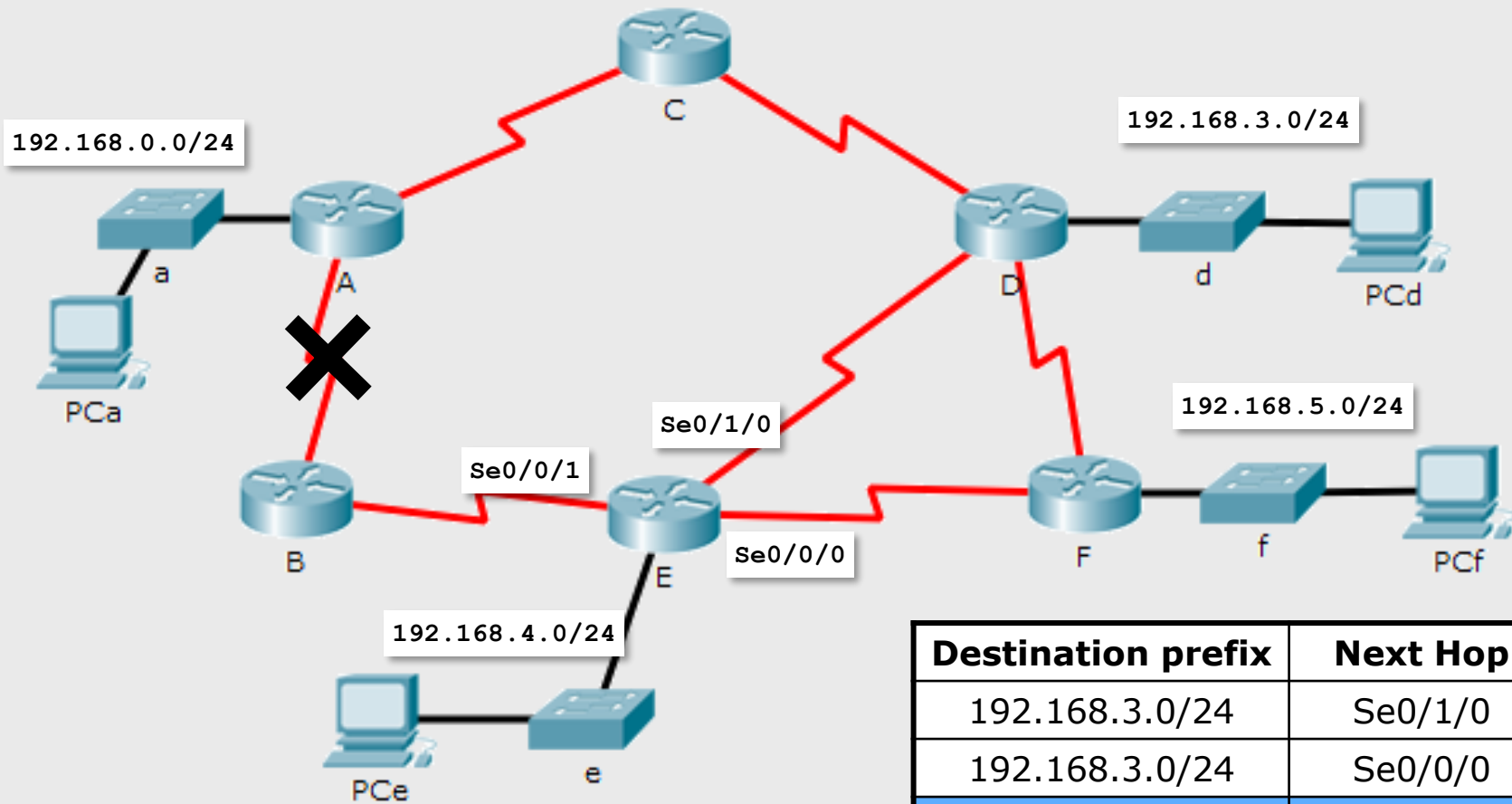
Dynamic routing
Link-state routing

Static routing limits



Destination prefix	Next Hop	Cost
192.168.3.0/24	Se0/1/0	30
192.168.3.0/24	Se0/0/0	40
192.168.0.0/24	Se0/0/1	30
192.168.0.0/24	Se0/1/0	40
192.168.5.0/24	Se0/0/0	30

Static routing limits



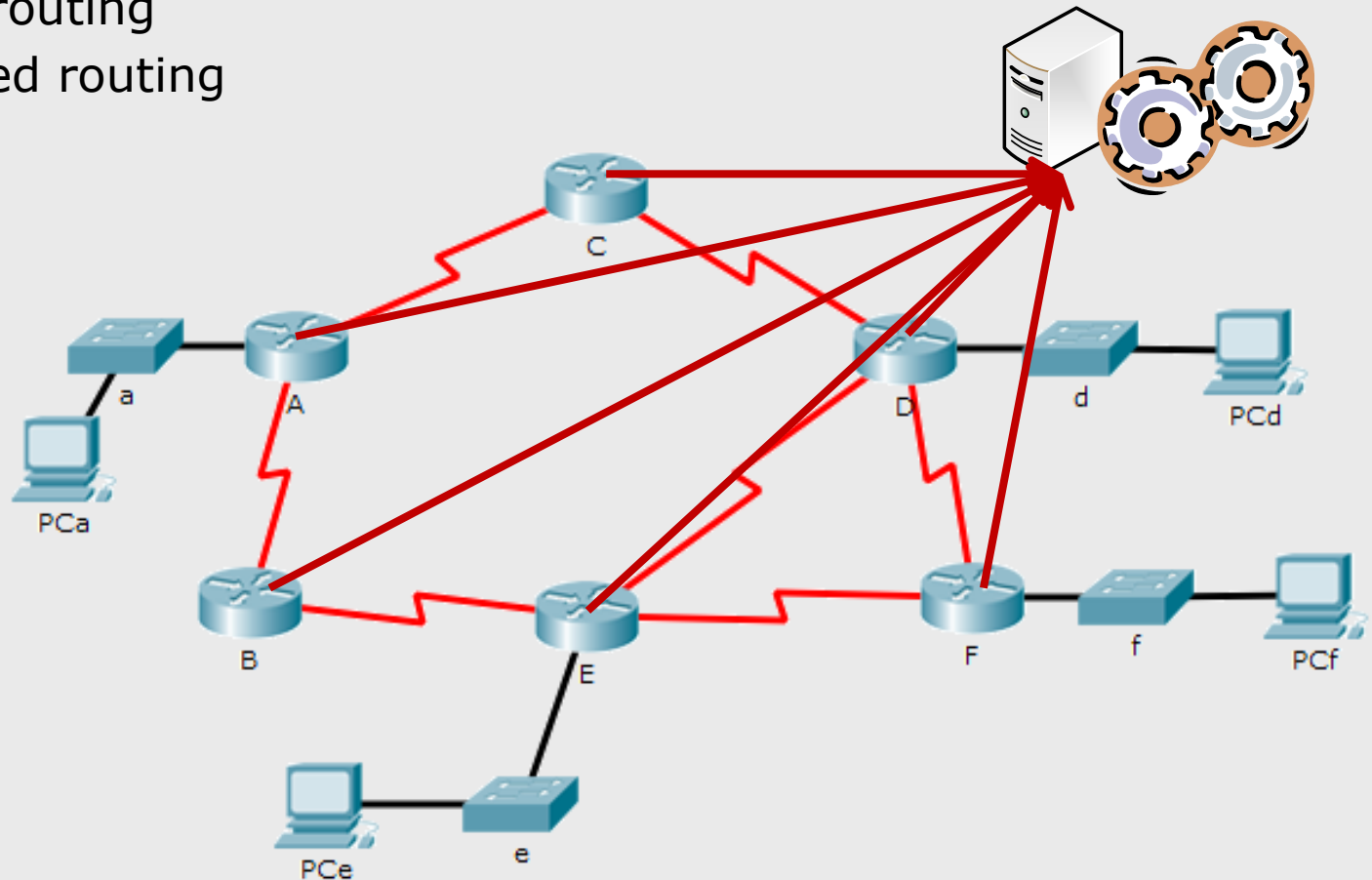
Destination prefix	Next Hop	Cost
192.168.3.0/24	Se0/1/0	30
192.168.3.0/24	Se0/0/0	40
192.168.0.0/24	Se0/0/1	30
192.168.0.0/24	Se0/1/0	40
192.168.5.0/24	Se0/0/0	30

Dynamic routing

- Routing tables are dynamically determined based on the computation done in the network according to a specific algorithm
 - Centralized routing
 - Isolated routing
 - Distributed routing

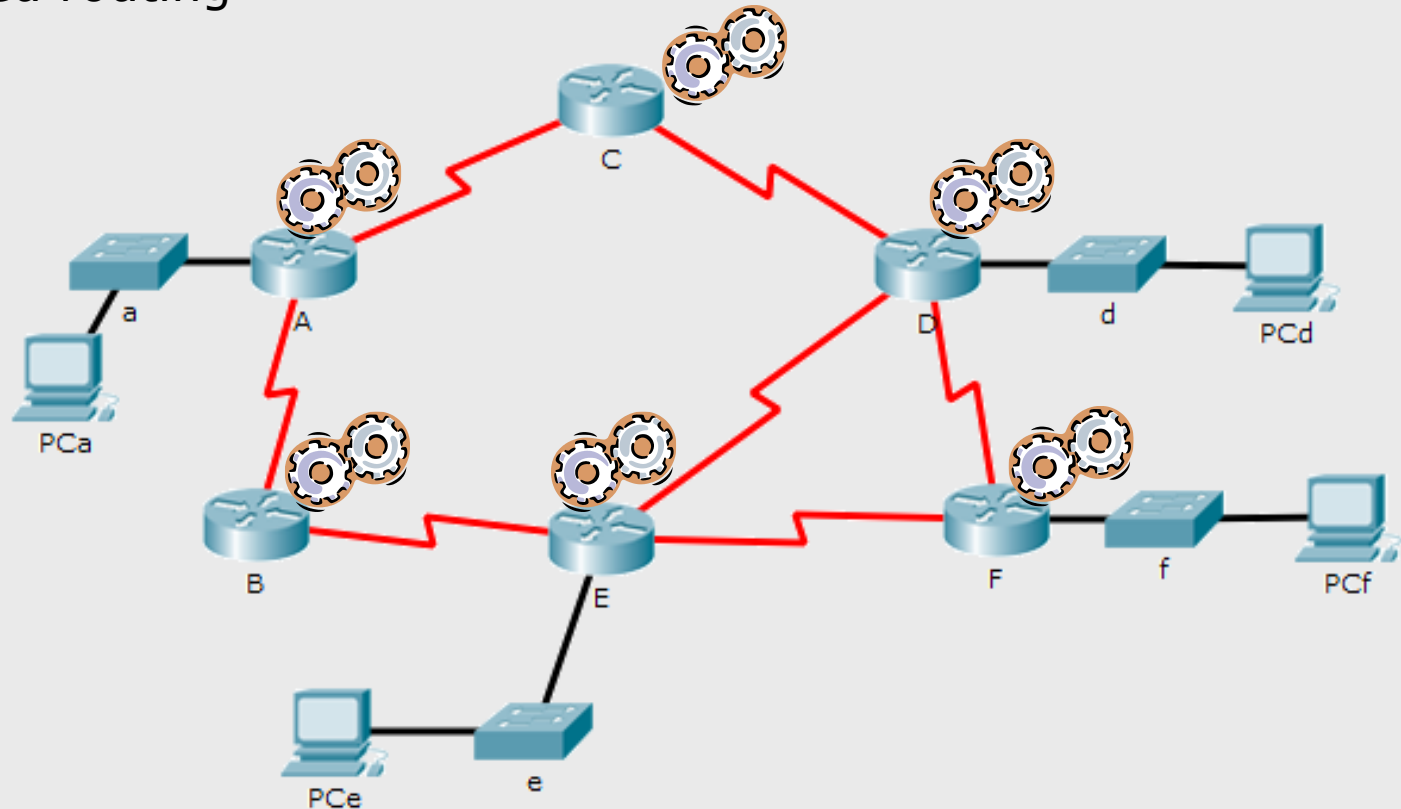
Centralized dynamic routing

- Routing tables are dynamically determined based on the computation done in the network according to a specific algorithm
 - **Centralized routing**: computation is done by a central unit and tables are then installed on all routers
 - Isolated routing
 - Distributed routing



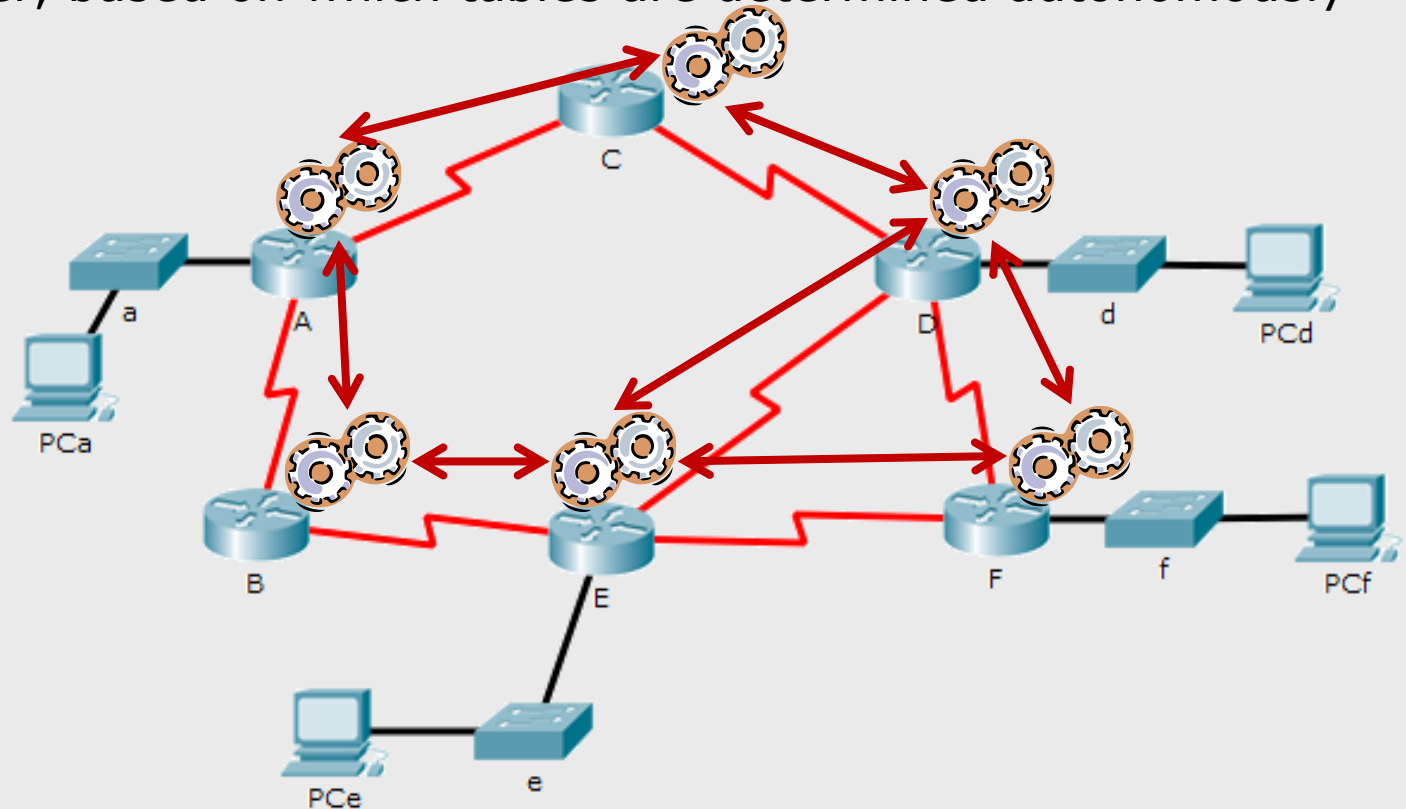
Isolated dynamic routing

- Routing tables are dynamically determined based on the computation done in the network according to a specific algorithm
 - Centralized routing
 - **Isolated routing**: each router builds its own table autonomously without interacting with other routers
 - Distributed routing



Distributed dynamic routing

- Routing tables are dynamically determined based on the computation done in the network according to a specific algorithm
 - Centralized routing
 - Isolated routing
 - **Distributed routing**: routers exchange topology information with each other, based on which tables are determined autonomously



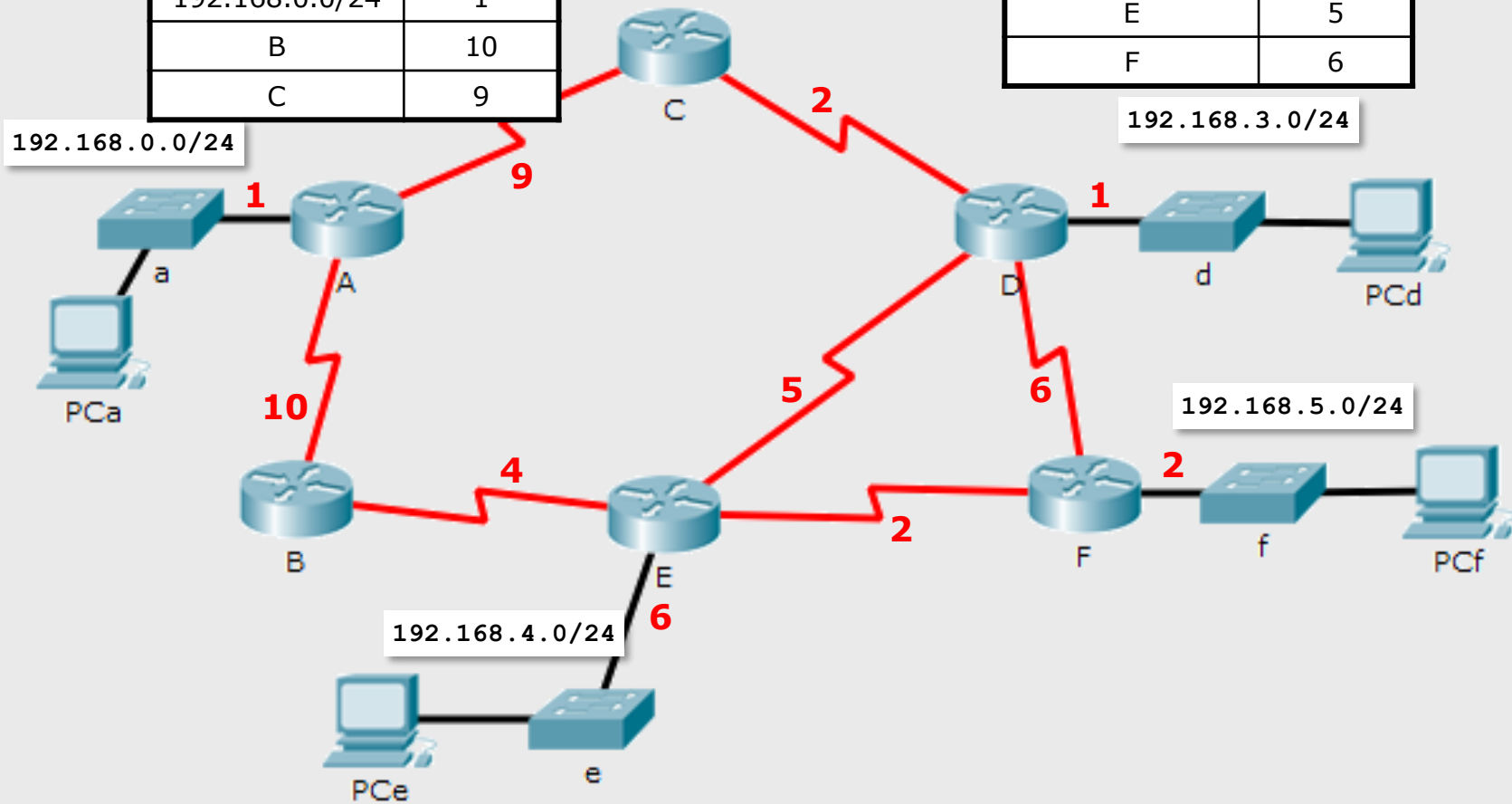
Recap – link state routing algorithm

- Each router advertises its neighbor topology, i.e., link states, through Link State Packets (LSP)
- For each link it is specified
 - The address of the neighboring node
 - The metric associated to traversing the link
- The router sends its own LSP to its neighbors, and cooperates to the propagation of LSPs generated by the other routers
- Each router receives all LSPs generated in the network
 - Stores LSPs in an LSP database
 - Determines the route topology from the LSP database
 - Computes the minimum cost path to each destination by means of the Shortest Path First algorithm
 - Builds the routing table based on the previous result

Recap – LSP propagation

Link	Metric
192.168.0.0/24	1
B	10
C	9

Link	Metric
192.168.3.0/24	1
C	2
E	5
F	6

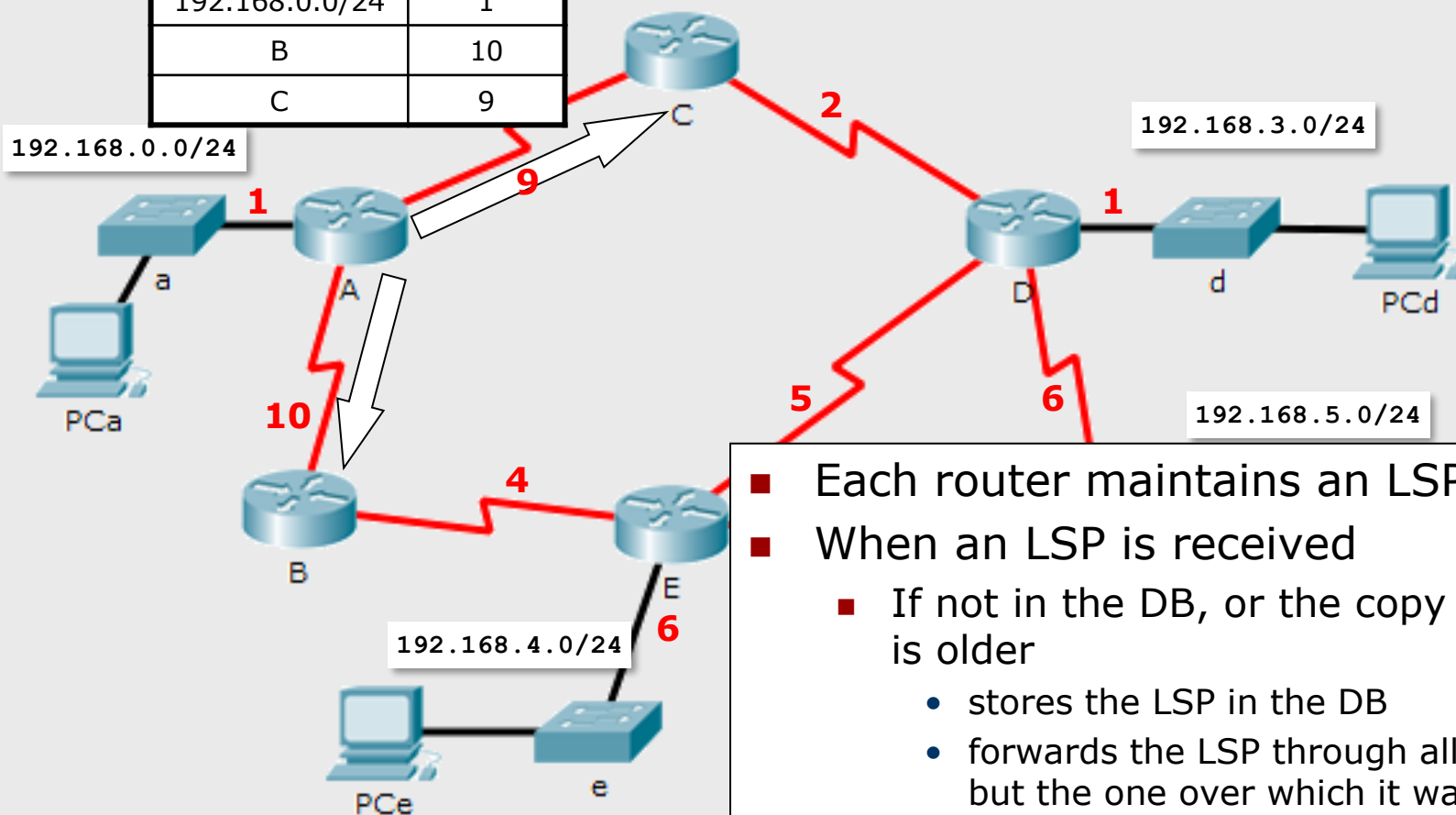


Selective flooding

Each LSP is propagated only the first time it is received

Recap – LSP propagation

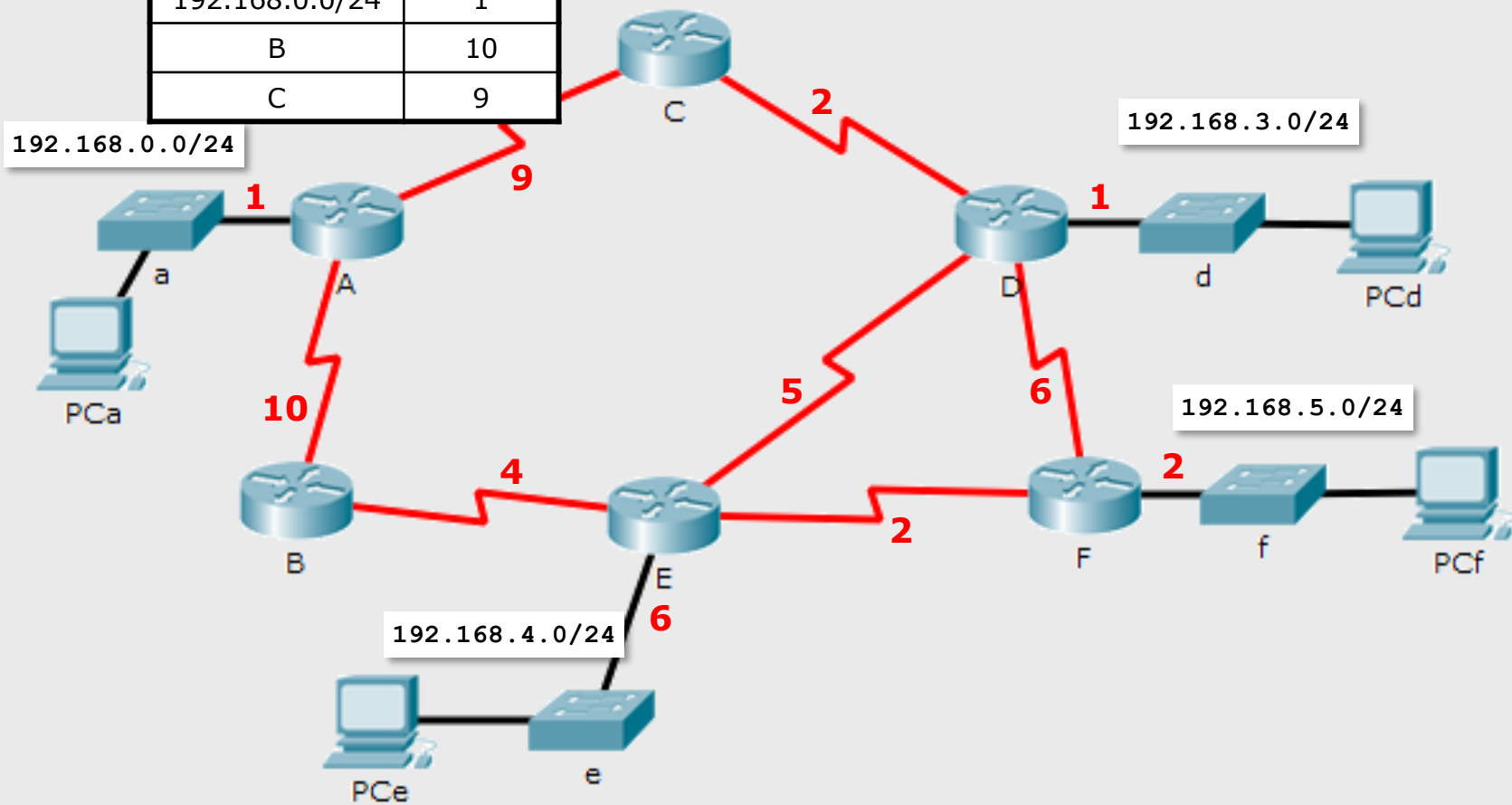
Link	Metric
192.168.0.0/24	1
B	10
C	9



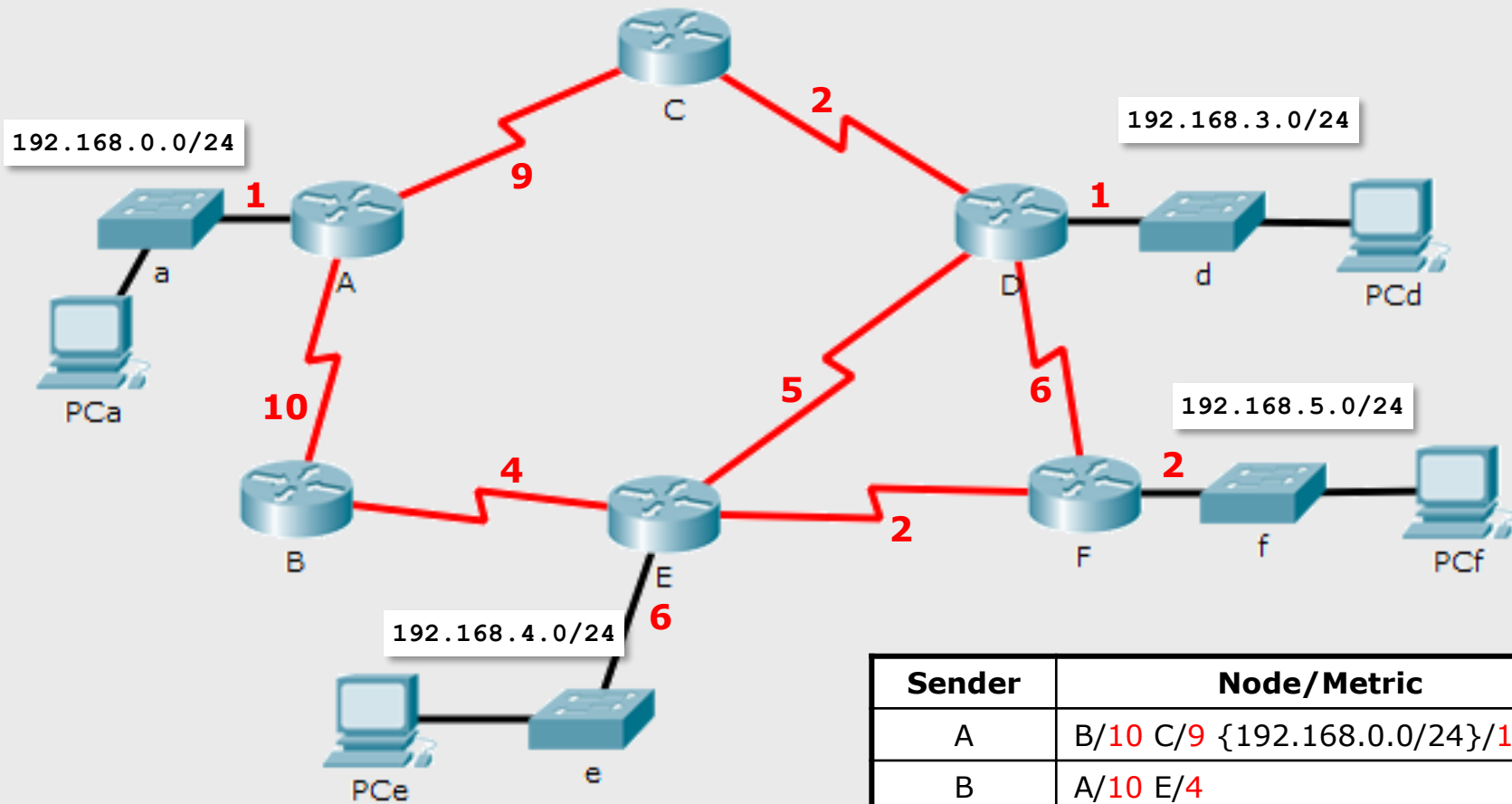
- Each router maintains an LSP DB
- When an LSP is received
 - If not in the DB, or the copy in the DB is older
 - stores the LSP in the DB
 - forwards the LSP through all interfaces but the one over which it was received
 - Otherwise, if the copy in the DB
 - is the same, ignores it
 - is more recent, sends back to the sender its own copy to update it

Recap – LSP propagation

Link	Metric
192.168.0.0/24	1
B	10
C	9



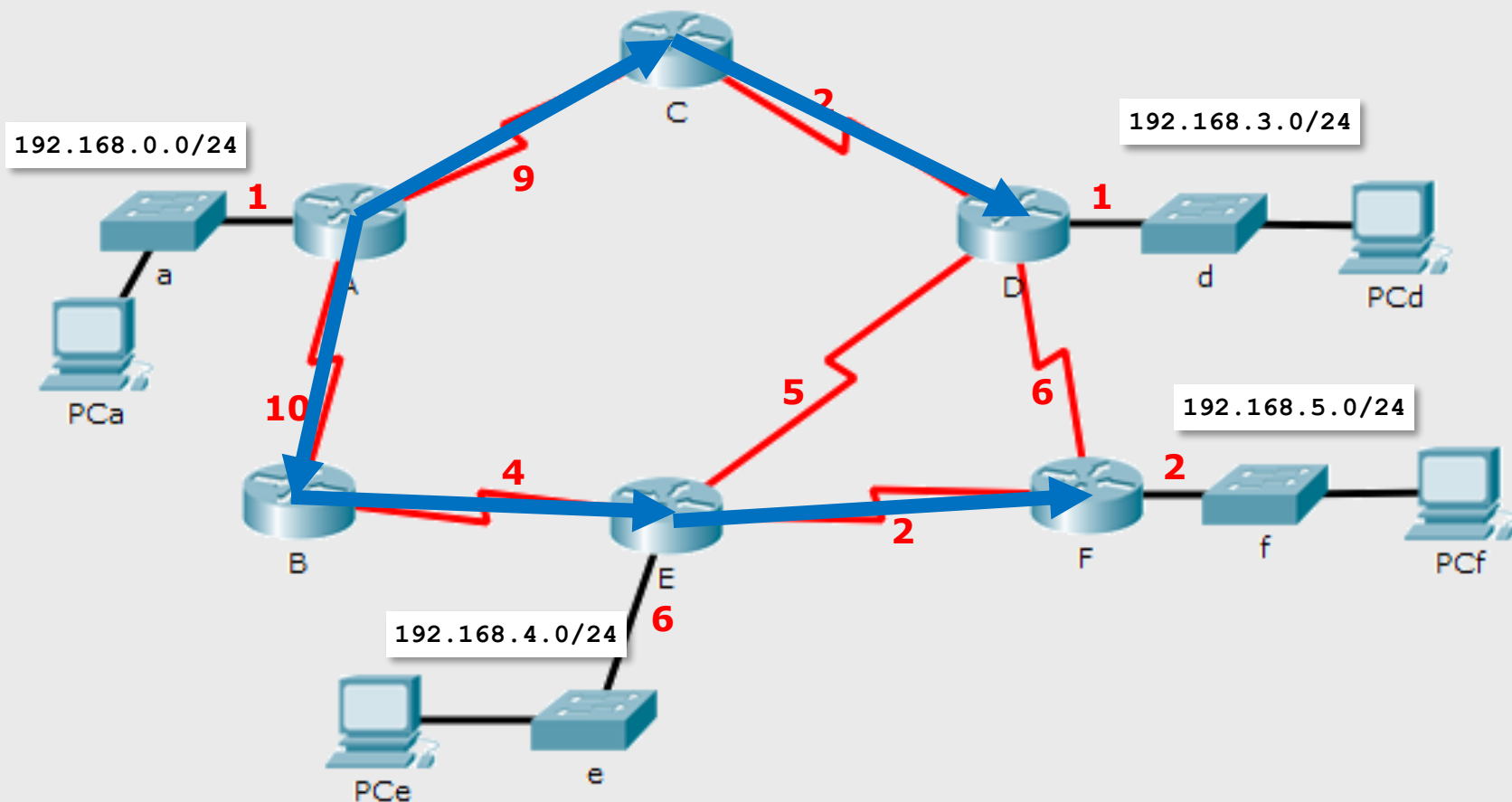
LSP Database



Sender	Node/Metric
A	B/ 10 C/ 9 {192.168.0.0/24}/ 1
B	A/ 10 E/ 4
C	A/ 9 D/ 2
D	C/ 2 E/ 5 F/ 6 {192.168.3.0/24}/ 1
E	B/ 4 D/ 5 F/ 2 {192.168.4.0/24}/ 6
F	D/ 6 E/ 2 {192.168.5.0/24}/ 2

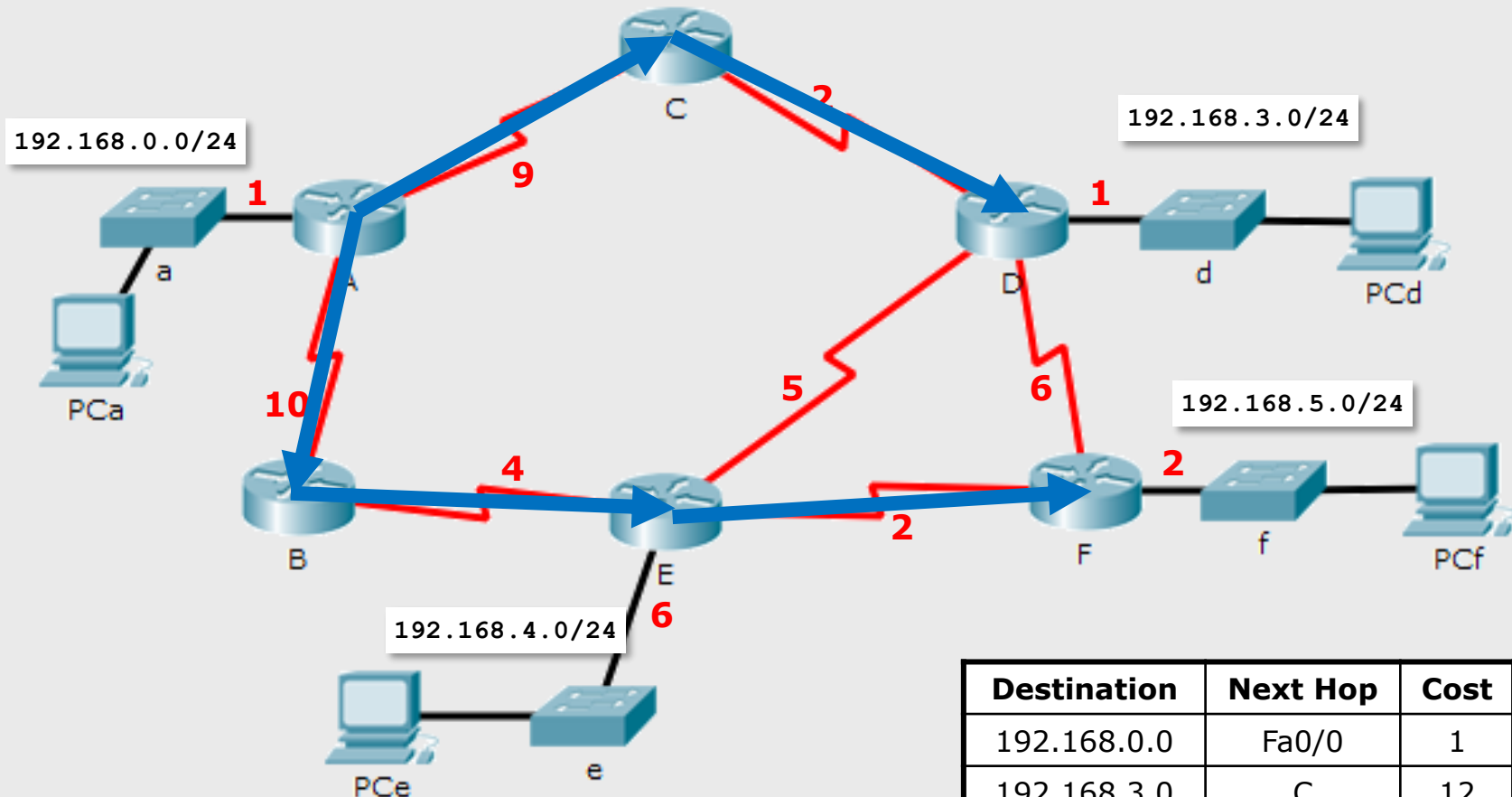
Dijkstra algorithm

- Shortest path tree rooted at A



Dijkstra algorithm

■ Shortest path tree rooted at A



Destination	Next Hop	Cost
192.168.0.0	Fa0/0	1
192.168.3.0	C	12
192.168.4.0	B	20
192.168.5.0	B	18

Proprietà dell'algoritmo link state

- Strong stability
 - Short convergence time
 - In the order of **$C \log(N)$** , where C is the number of links and N the number of nodes
 - Routers compute the topology independently of each other (no bottleneck)
 - Quick reaction to topology changes
 - Little inclination to create loops
 - Depends on the coherence property of shortest path trees
 - Temporary loops only during LSP exchange, which is very fast
- Easy troubleshooting, based on the availability of the LSP database at each router